

Do students' perceptions of their weight reflect their actual BMI percentile?

12,853

STUDENTS ANALYSED
14,041 raw - 1,188 cleaned

GROUPS · EXPLANATORY categorical · 5 levels

PerceptionOfWeight

1	Very underweight	n 207
2	Slightly underweight	n 1,436
3	About the right weight	n 7,351
4	Slightly overweight	n 3,318
5	Very overweight	n 541

RESPONSE VARIABLE

BMIPCT BMI percentile

Quantitative measure, ranges 0–100. Higher = heavier relative to peers.

METHOD $\alpha = 0.05$

One-way ANOVA + Tukey HSD post-hoc

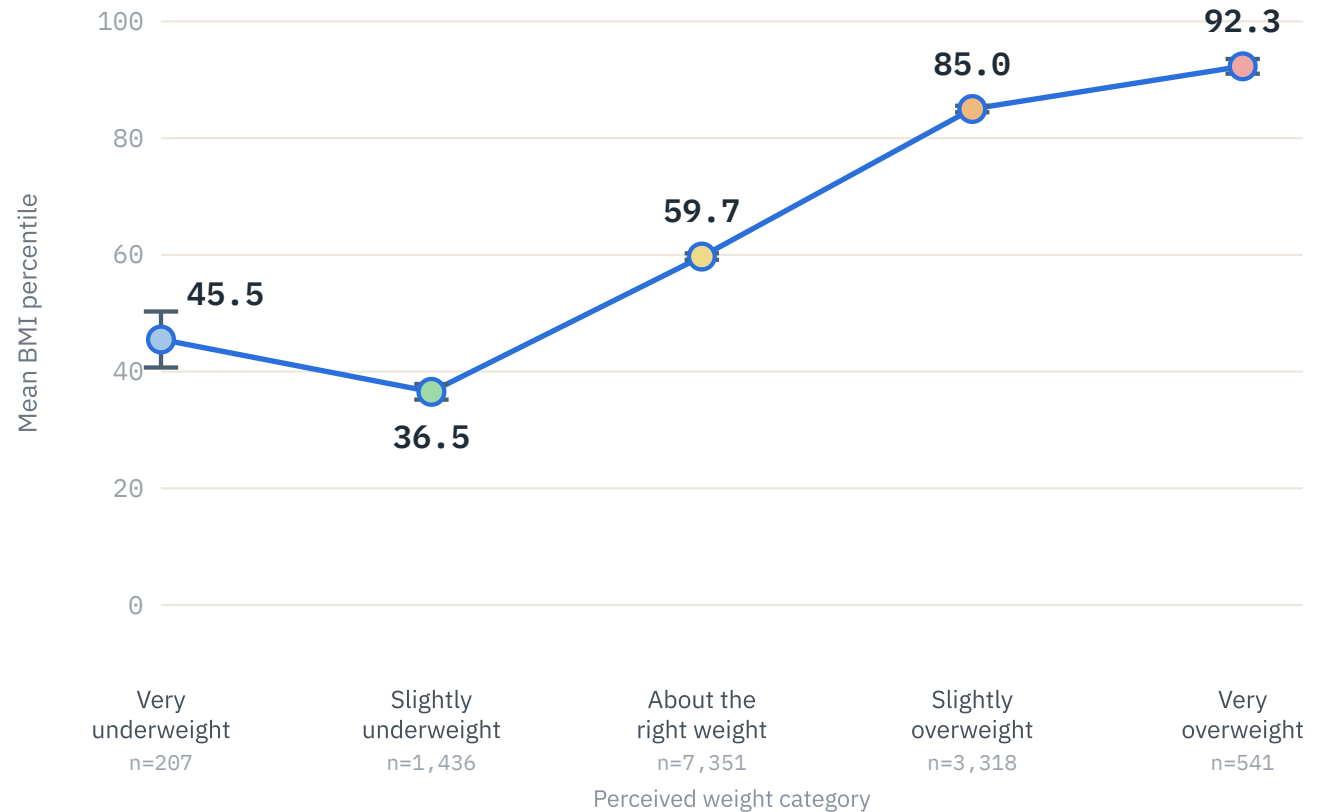
Let μ_k = population mean BMIPCT of perceived-weight group k (1–5).

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

$$H_a: \mu_i \neq \mu_j \text{ for at least one pair } (i, j)$$

Mean BMI percentile is generally higher as perceived weight increases

95% CI error bars



KEY RESULT · ANOVA

F = 1,557

$p < 0.0001 \rightarrow$ Reject H_0

10/10

TUKEY PAIRS

Every pairwise comparison is significant ($p < .0001$). Largest gap: Very overweight sits **+55.8 pct** above Slightly underweight.

CONCLUSION

Students' perceived weight is clearly **associated** with their actual BMI percentile – those who feel heavier do have higher percentiles. Because YRBS is observational survey data, this is association, not causation.